



Ramakrishna Mission Vivekananda Educational and Research Institute

PO Belur Math, Howrah, West Bengal 711 202

School of Mathematical Sciences

Department of Computer Science

MSc BDA : Batch 2022-24, Semester I, Final Exam

DA102: Basic Statistics

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Student Name (in block letters):

Date: 25 January 2023

Student Roll No:

Max Marks: 100

Signature:

Time: 3hrs

Answers must be properly justified to deserve full credits.

1. (10 points)

Choose the correct answer.

- (a) (1 point) In a Pie calculate the angles for each sectors by the following formula
1. $\frac{\text{componentPart}}{\text{Total}} \times 100$; 2. $\frac{\text{componentPart}}{\text{Total}} \times \pi$; 3. $\frac{\text{Total}}{\text{componentPart}} \times 360^\circ$; 4. $\frac{\text{componentPart}}{\text{Total}} \times 360^\circ$.
- (b) (1 point) Cumulative frequency polygon can be used for the calculation of
1. Arithmetic mean; 2. Median; 3. Mode; 4. Geometric mean.
- (c) (1 point) A Histogram contains a set of
1. Adjacent rectangles; 2. Non Adjacent Rectangles; 3. Adjacent squares; 4. Adjacent triangles.
- (d) (1 point) Which of the following is not a one-dimensional diagram
1. Simple bar diagram; 2. Multiple bar diagram; 3. Component bar diagram; 4. Pie diagram.
- (e) (1 point) A distribution in which the observations are concentrated at one end of the distribution is called a
1. Symmetric distribution; 2. Normal distribution; 3. Skewed distribution; 4. Uniform distribution.
- (f) (1 point) Component bar charts are used when data is divided into
1. parts; 2. groups; 3. circles; 4. none of these.
- (g) (1 point) Histogram can be drawn only for
1. Discrete frequency distribution; 2. Cumulative frequency distribution 3. Continuous frequency distribution; 4. Relative frequency distribution.
- (h) (1 point) In a histogram the area of each rectangle is proportional to
1. the class mark of the corresponding class interval; 2. the class size of the corresponding class interval;
3. frequency of the corresponding class interval; 4. cumulative frequency of the corresponding class interval.
- (i) (1 point) A frequency polygon is constructed by plotting frequency of the class interval and the
1. upper limit of the class; 2. lower limit of the class; 3. mid value of the class; 4. any values of the class.
- (j) (1 point) Frequency polygon can be drawn with the help of
1. Historigram; 2. Histogram; 3. Circle; 4. Percentage.

2. (18 points)

For a series data $\{x_1, \dots, x_n\}$, prove that

$$\frac{R^2}{2n} \leq \sigma^2 \leq \frac{R^2}{4},$$

where σ^2 and R are the variance and the range of the data, respectively.

3. (18 points)

If $X_1, \dots, X_n$ is a sample of size n (chosen without replacement) from a population $\{U_1, U_2, \dots, U_N\}$, with mean μ and variance σ^2 , then show that

$$E[\bar{X}] = \mu \text{ and } Var[\bar{X}] = \frac{\sigma^2}{n} \left(\frac{N-n}{N-1} \right),$$

where $\bar{X} = (X_1 + \dots + X_n)/n$

4. (18 points)

Let $X_1, \dots, X_n$ be independent random variables with densities

$$f_{X_i}(x|\theta) = \begin{cases} e^{i\theta-x}, & x \geq i\theta \\ 0, & x < i\theta \end{cases}.$$

Prove that $T = \min_i(X_i/i)$ is a sufficient statistic for θ .

5. (18 points)

One observation is taken on a discrete random variable X with pmf $f(x|\theta)$, where $\theta \in \{1, 2, 3\}$. Find the MLE of θ .

x	$f(x 1)$	$f(x 2)$	$f(x 3)$
0	1/3	1/4	0
1	1/3	1/4	0
2	0	1/4	1/4
3	1/6	1/4	1/2
4	1/6	0	1/4

6. (18 points)

Let $X_1, \dots, X_n$ be iid $Bernoulli(p)$. Show that the variance of $\bar{X}$ attains the Cramer-Rao Lower Bound and hence $\bar{X}$ is the best unbiased estimator of p .

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This exam has total 6 questions, for a total of 100 points and 0 bonus points.
